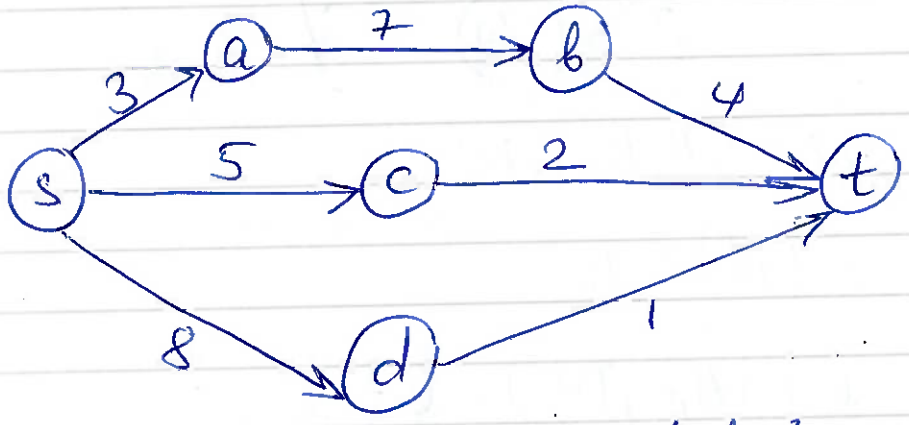
week 4 09.02.2026

① slides

② network flow game example



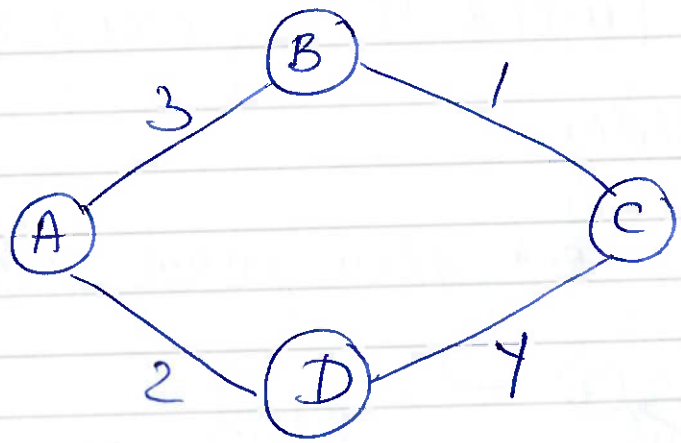
$N = \{sa, ab, bt, sc, ct, sd, dt\}$

$v(C = \{sa, ab, bt\}) = \min\{c_{sa}, c_{ab}, c_{bt}\} = 3$

$v(D = \{sc, ct\}) = \min\{c_{sc}, c_{ct}\} = 2$

$v(N) = 6$

③ Induced subgraph game example



$N = \{A, B, C, D\}$

$v(N) = 10$

$v(\{A, C, D\}) = 6$

(4) Shapley value of induced subgraph games

Consider an induced subgraph game G given by a graph $G = (N, E)$ and a weight vector w, and let $\{e_1, \dots, e_m\}$ be the set of edges of this graph. We can decompose the graph G into m graphs $G^1, \dots, G^m$,

where for every $j=1,\dots,m$ the graph G^j has the vertex set N and a single edge e_j (with the same weight as in the original graph); let G^j denote the game that corresponds to the j-th graph in this sequence. It is easy to see that

$$G(G,w) = G^1 + \dots + G^m$$

Therefore, by additivity axiom for each player $i \in N$ we have

$$\varphi_i(G) = \sum_{j=1}^{m} \varphi_i(G^j),$$

and it remains to compute the Shapley value of player i in the game G^j.

Suppose first that is <u>not</u> incident to e_j. Then i is clearly a dummy in G^j

and therefore we have $\varphi_i(G^j)=0$. On the other hand, suppose that $e_j=(i,l)$ for some $l \in N$. Then i and l are symmetric in G^j. Since the value of the grand coalition in G^j equals w_{il}, by efficiency and symmetry axioms we get $\varphi_i(G^j)=\frac{w_{il}}{2}$.

Putting these observations together, we conclude that an agent's Shapley value is <u>half</u> the income from the edges in the graph to which it is attached:

$$\varphi_i(G)=\frac{1}{2}\sum_{(i,j)\in E} w_{ij}$$

5 Shapley value applications: feature attribution

The main application of Shapley value in AI is in a hot field called inter-

pretability, where we want to understand how black-box models make predictions or computations.

The Shapley value was a concept originally known only within cooperative game theory, until it gained widespread popularity in the AI community following a 2017 NeurIPS paper. It has since become one of the most well-known and widely used methods for interpreting complex models.

Here is an example of how the Shapley value is useful for interpretability:

Given a machine learning model $f: \mathbb{R}^d \to \mathbb{R}$ and a point $x \in \mathbb{R}^d$, what is the influence of each feacher

in x on the model's prediction $f(x)$?

For example, we might have a model trained to predict whether individuals have income greater than $50K based on U.S. census data. The input features might include age, education level, marital status, and hours worked per week. We want to quantify the contribution of each feature to the model's output.

The Shapley value allows us to assign a numerical value to each feature's contribution, capturing how much each one "pushes" the model's prediction up or down. These contributions are computed by considering all possible coalitions (subsets of features) and averaging marginal contributions, similar

to cooperative game theory.

We can plot shapley values along a line, while also specifying a baseline that represents the model's prediction if no features are known (i.e., the expected output over all data points). Each feature then either pushes the prediction to the right (increasing the model's output) or to the left (decreasing it).
Items that fall to the left of the baseline represent features that increase the prediction $f(x)$, or push the prediction toward the right. Items that fall to the right of the baseline represent features that decrease the prediction, or push it toward the left.
The further to the right or left a feature pushes, the

more influential it is in the model's final decision. Thus, this kind of visualization gives a clear and intuitive sense of which feature matter most and in which direction they are influencing the output. f here could have been arbitrarily complicated, but this Shapley value representation gives us insight into the impacts of the features.

Given a model $f: \mathbb{R}^d \to \mathbb{R}$ and a point x, define a cooperative game:

- The players are the d features
- $v(S) = f(x_S)$, where x_S is x with the features in $N \setminus S$ "removed"

Now, we can compute the Shapley value of the features.